



Pharm*Achieve*

PRURITUS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACE-I Angiotensin-Converting Enzyme Inhibitor

ADE Adverse Drug Event

AE Adverse Effect

BB Beta-Blocker

BID Bis In Die (Twice Daily)

CBC Complete Blood Count

CCB Calcium Channel Blocker

EGFR-I Epidermal Growth Factor Inhibitor

G-CSF Granulocyte-Colony Stimulating Factor

H1 Histamine 1

HIV Human Immunodeficiency Virus

IV Intravenous

NHP Natural Health Product

PO Per Os (Oral)

PRN Pro Re Nata (As Needed)

Q Quaque (Every)

RCT Randomized Controlled Trial

SSRI Selective Serotonin Reuptake Inhibitor

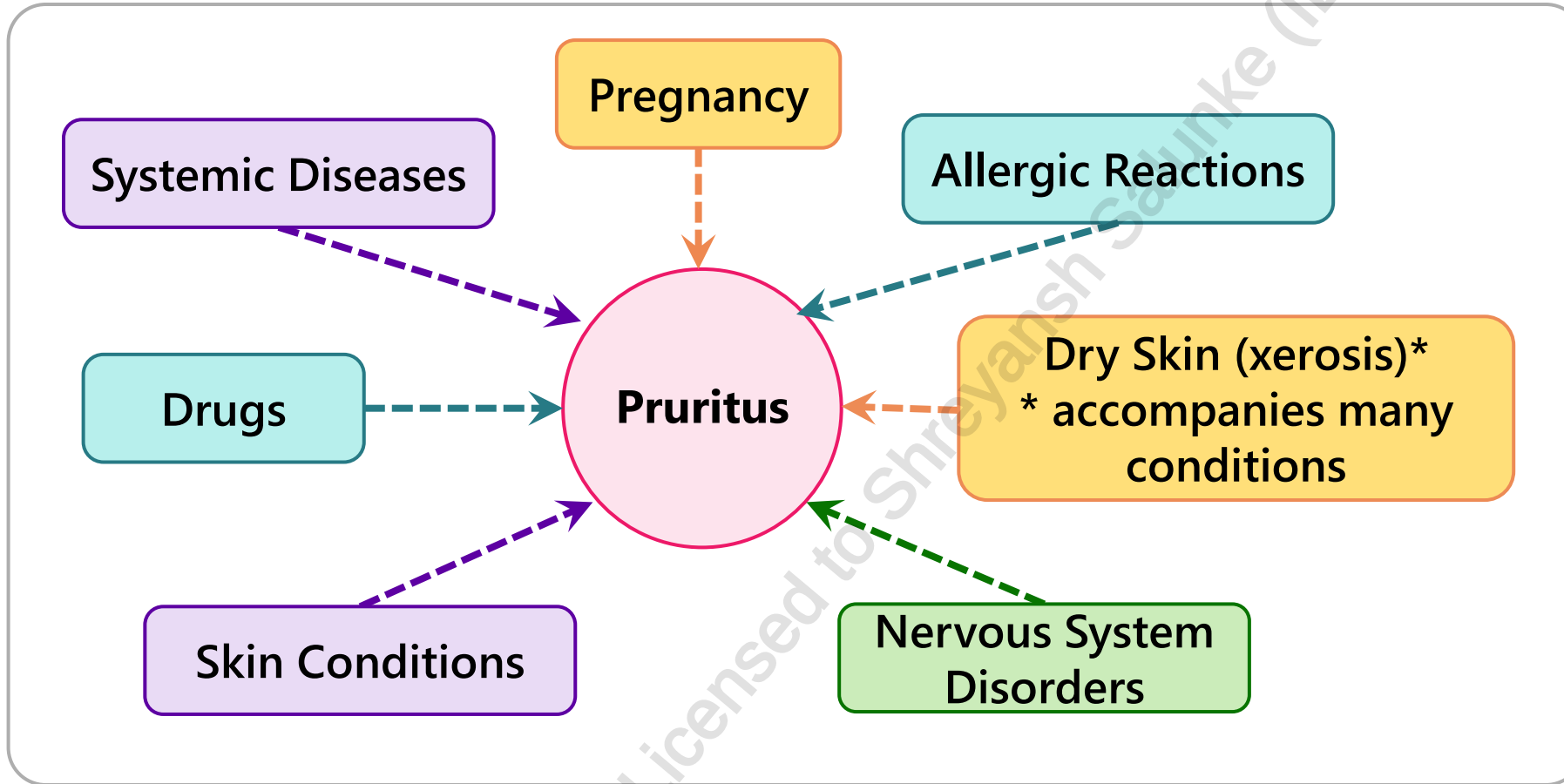
TMP-SMX Trimethoprim-Sulfamethoxazole

OBJECTIVES

- 1 Explain the pathophysiology of pruritus
- 2 Identify the clinical presentation and diagnostic method
- 3 Identify goals of therapy and monitoring parameters
- 4 Understand the management of pruritus including non-pharmacological and pharmacological alternatives

DEFINITIONS AND PATHOPHYSIOLOGY

Pruritus is defined as **itchy skin** that may stem from an underlying cause or condition



CLINICAL PRESENTATION

- Itching
- Redness
- Dryness/cracked skin
- Bumps
- Blisters
- Leathery/scaly texture

RISK FACTORS

Chronic renal disease

Chronic liver disease

Atopy

Seasonal/
environmental allergies

DIAGNOSIS

Patient history

- Location/duration/severity of symptoms
- Aggravating and relieving factors
- History of systemic diseases
- Skin care and cleansing routine
- Fever, chills and night sweats; unintended weight loss
- Prescription/non-prescription medications, NHPs and illicit drug use
- Change in products that touch skin (soap, laundry detergent, etc.)

Physical examination

- Skin rash, dryness, burrows, excoriations, dermatographism, abnormal skin pigmentation, lymphadenopathy, hepatosplenomegaly

Laboratory investigations

- Required if a primary dermatologic cause has been ruled out
- CBC with differential, fasting serum glucose levels, renal function tests, liver function tests (including bilirubin), thyroid function tests, chest x-ray and HIV tests

DRUG-RELATED CAUSES

- Antibiotics
 - Penicillins, TMP-SMX, quinolones
- Heparin
- ACE-Is, CCBs, BBs, hydrochlorothiazide
- Statins
- Opioids
- Chloroquine
- Hydroxyethyl starch (12-54% of patients)
 - Used in IV therapy
- Epidermal growth factor inhibitors (EGFR-I)
 - Used in cancer treatments (e.g. gefitinib)
- Granulocyte colony stimulating factor (G-CSF)
- Interleukin-2

GOALS OF THERAPY

- ✓ Determine underlying cause
- ✓ Rule out possible underlying systemic diseases
- ✓ Individualize therapy to maximize efficacy and minimize adverse effects

NON-PHARMACOLOGICAL THERAPY

Avoid aggravating factors

- **Avoid excessive exercise, humidity, spicy foods, caffeine, alcohol, high temperatures**
- Use a flare diary to identify aggravating factors

Break itch-scratch cycle

- Cool compresses, colloidal oatmeal or baking soda bath, trim fingernails

Minimize friction and skin irritation

- Wear soft and loose clothing, and avoid woolen and synthetic fabrics
- Use fragrance-free products for laundry and personal care

Skin hydration

- Use gentle cleansers and mild, fragrance-free soap with neutral pH (avoid alkaline soaps)
- Moisturization for all patients
- **Avoid hot water and hot tubs**

Ensure adequate sleep

- Pruritus worsens at night: apply moisturizer, emollient or antipruritic cream before bedtime
- Keep the bedroom at a cool ambient temperature and use light bed cloths

Behavioural therapy, relaxation, stress reduction

PHARMACOLOGICAL THERAPY - TOPICAL ANTIPRURITICS

- Topical therapies are 1st line for patients with mild or localized itching
- Avoid topical benzocaine due to its sensitizing properties

Menthol or camphor

- Light, non-perfumed lotions containing menthol or camphor PRN up to TID can be used (concentration varies)
 - Reduces itching sensations by eliciting a cooling sensation
 - Menthol is less toxic when absorbed systemically

Zinc oxide

- Zinc oxide (10-50%) may be used for localized pruritus

Pramoxine

- Consider pramoxine (1%), a low sensitizing topical anesthetic

Calamine

- Has astringent properties that may be useful (e.g. in patients with chickenpox). However, it will excessively dry out the skin if used long-term

PHARMACOLOGICAL THERAPY - TOPICAL ANTIPRURITICS

Topical corticosteroids

- Topical corticosteroids are useful if the pruritus is the result of an inflammatory skin condition (redness with itching and/or tenderness)
- Should not be used for chronic, generalized pruritus

Tacrolimus and pimecrolimus

- Topical immune modulators may be useful

Capsaicin

- Consider capsaicin as it blocks the transmission of itching along the peripheral nerves
- May cause a **local burning sensation** and redness when initially applied
- Requires regular use for effectiveness

PHARMACOLOGICAL THERAPY - SYSTEMIC ANTIHISTAMINES

- Useful for urticaria, but little impact on other conditions associated with pruritus
- First-generation antihistamines (e.g. diphenhydramine, hydroxyzine) can be useful for sleep
 - Concerns of impaired daytime performance, anticholinergic ADEs
- First-line for urticaria: Histamine 1 (H_1) receptor blockers
 - More effective when used to prevent histamine release, thus administer in regularly scheduled doses in chronic urticaria
 - **Second-generation antihistamines** are preferred for their favourable side effect profile (e.g. non-sedating)
 - If inadequate response, other drugs with antihistaminic properties can be tried (e.g. doxepin)
 - Combining with **leukotriene antagonists** may be helpful (**data is inconsistent**)



Histamine 2 (H_2) receptors are not directly involved in the symptoms of itching

PHARMACOLOGICAL THERAPY - OTHER SYSTEMIC AGENTS

Symptom improvement with antidepressants may take up to 4 weeks to be seen; many of the therapies have limited evidence available or have results that have not gone through large RCTs to be confirmed

Doxepin

- Tricyclic antidepressant with antihistaminic properties
- Can be useful in various aetiologies such as chronic urticaria and psychogenic causes

Mirtazapine

- Antidepressant with antihistaminic and serotonergic properties
- Can be effective in pruritus from various aetiologies

Paroxetine, sertraline and fluvoxamine (SSRIs)

- May help with itch related to chronic kidney disease, malignancies and cholestasis

PHARMACOLOGICAL THERAPY - OTHER SYSTEMIC AGENTS

Gabapentin and pregabalin

- Inhibits central itching pathways
- Used in uremic pruritus (level I evidence^{*}) or neuropathic-associated pruritus (level II evidence^{**})

Naltrexone and bupropion (level I evidence^{*})

- Endogenous opioid pathway imbalances may contribute to the pathophysiology of pruritus
- Naltrexone is a μ -receptor antagonist shown to improve pruritus in patients with cholestasis, atopic dermatitis, end-stage renal disease and burns. Negative effects have also been reported.
- Bupropion is a κ -receptor agonist and mixed μ -receptor competitive antagonist shown to improve symptoms in patients with intractable pruritus
- Difelikefalin is a κ -receptor agonist indicated for the treatment of moderate to severe pruritus in patients with CKD on hemodialysis

^{*} Level I evidence requires at least 1 properly conducted RCT, systematic review, or meta-analysis

^{**} Level II evidence includes other comparison trials, non-randomized, cohort, case-control, or epidemiologic studies, and preferably more than 1 study

PHARMACOLOGICAL THERAPY - SYSTEMIC AGENTS

FYI

Ketotifen

- Selective H1 antihistamine that stabilizes mast cells and inhibits mediator release
- Small studies have shown benefit in urticaria and mastocytosis

Montelukast

- Leukotriene receptor antagonist
- Used as adjunctive therapy to standard antihistamines for patients with chronic idiopathic urticaria (especially severe)
- May also be effective in uremic pruritus

PHARMACOLOGICAL THERAPY - SYSTEMIC AGENTS

FYI

Cholestyramine and colestipol

- Used for pruritus related to cholestatic liver disease, although good evidence is lacking
- Consider colesevelam as alternative

Rifampin

- Effective and safe when used in patients with pruritus from chronic cholestasis
- Lowers hepatocyte bile salt concentrations

Serotonin antagonists

- Ondansetron has been used to treat pruritus of hepatic or renal origin
 - 2 systematic reviews found the benefits to be negligible

Aprepitant (substance P antagonist)

- Case reports and small studies show effectiveness in refractory and severe pruritus associated with some cancers and biologic cancer treatments

Systemic corticosteroids

- Only if there is cutaneous inflammation with pruritus and other therapies are inappropriate

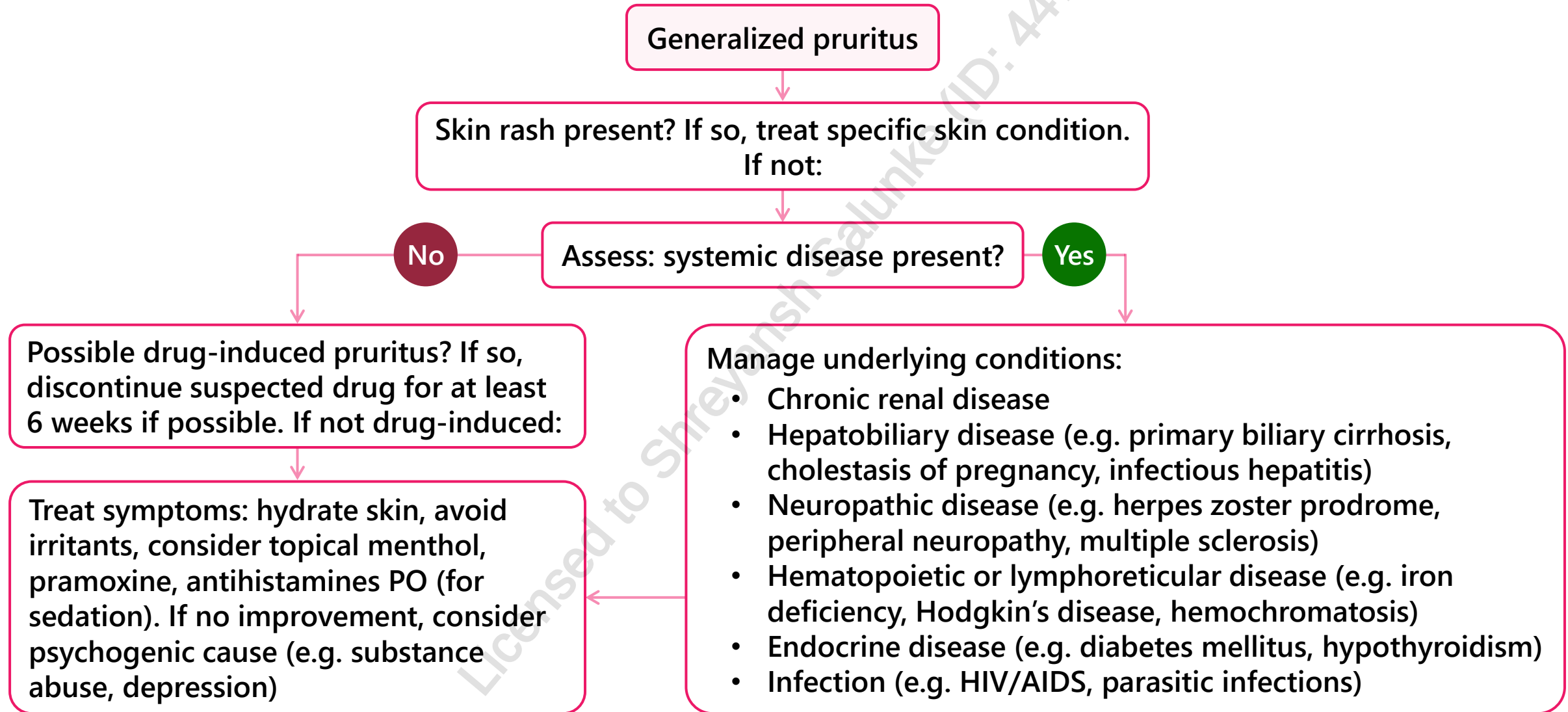
COUNSELLING TIPS ON OVER-THE-COUNTER PRODUCTS

Class	Drug example	Common side effects	Comments
Antihistamines	Diphenhydramine	Dizziness, drowsiness, fatigue, rash, dry mouth, blurred vision. Occasionally may cause paradoxical excitation in children	Indicated only in urticaria (histamine-mediated) Often used for sedative properties in other types of pruritus Liquid form available
Astringents	Calamine lotion	Skin irritation	Use only on moist lesions, otherwise too drying
Bath Emollients	Colloidal oatmeal (Aveeno® Bath Preparations, generics)		Add to warm (not hot) bath water as needed – soak for 10–15 min
Counterirritants	Menthol	Skin irritation (burning, redness) at higher concentrations	Lower concentration products preferred Do not apply to large areas of skin or open wounds

COUNSELLING TIPS ON OVER-THE-COUNTER PRODUCTS

Class	Drug example	Common side effects	Comments
Counterirritants	Camphor	Skin irritation (burning, redness) at higher concentrations. Absorbed through intact and broken skin Excessive topical use can result in systemic toxicity (including nausea, vomiting, headache, dizziness, tremors, seizures)	Lower concentration products preferred. Do not apply to large areas of skin or open wounds. Young children may be more susceptible to adverse effects associated with even minor systemic absorption of camphor (including via topical use and inhalation); the American Academy of Pediatrics recommends that camphor not be used in children
Local Anesthetics	Pramoxine (Gold Bond® Anti-Itch Lotion)	Contact dermatitis, burning, stinging	Do not apply to large areas of skin or open wounds
Topical Corticosteroids	Hyderm® (hydrocortisone) 0.5% or 1% cream or ointment	Local burning, irritation, dryness	Indicated only for pruritus due to cutaneous inflammation (visible redness, itch and/or tenderness) Do not use for long-term generalized pruritus Use not recommended on face, groin, axillae or open skin lesions

MANAGEMENT OF PRURITUS



MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Pruritus, redness, dryness	Improve/resolve	Days to weeks (depending on severity, underlying cause, and treatment of choice)
Secondary infections	Prevent	

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Local skin reactions e.g. burning, tingling, contact dermatitis	Prevent/minimize	Duration of therapy
AEs of systemic therapy e.g. nausea, vomiting, dizziness, drowsiness		

TAKEAWAY

- Avoid further drying of the skin: use gentle cleansers such as synthetic detergents instead of soap, moisturize diligently, and minimize hot water showers/baths
- Do not use topical corticosteroids or immune modulators unless skin inflammation is present
- Evaluate for burrows to monitor for scabies
- Assess for dermatographism or pressure sensitivities
 - Consider regular daily use of non-sedating antihistamines with periodic tapering
- Topical medications can be kept in a refrigerator as physical cooling increases the antipruritic effect of the medications

CASE STUDY

EK is a 45-year-old female who presents to your pharmacy asking about hydrocortisone cream. When you ask her what she is using it for, she tells you that she's been experiencing itchy skin since she returned from her vacation in Italy three weeks ago. At first, it was just a mild itch on her hands, then it spread to her arms and feet, and now with the humid, hot weather, she's starting to feel it all over her back as well. The itchiness is sometimes so severe it interrupts her sleep. She tried a baking soda bath that was recommended by her family physician, and it provided temporary relief, but the itchiness returned shortly after. She heard about hydrocortisone cream from her neighbour who has been using it for many years with excellent results, so she would like to try it out. EK does not have any environmental, medical or food allergies. Her medical conditions include type 2 diabetes mellitus, for which she takes metformin 1000 mg PO BID, and chronic spinal neuropathic pain, for which she uses TEVA-LENOLTEC No. 3 (acetaminophen 300 mg/caffeine 15 mg/codeine 15 mg) 1-2 tablets PO Q6-8H PRN.

1. All of the following are important counselling points for EK EXCEPT:
 - a) Hydrocortisone cream is a good option for EK's itch due to its mild steroid potency
 - b) Topical corticosteroids are not recommended to be used long term and over large body surface areas
 - c) EK might benefit more from a systemic intervention due to the widespread nature of the rash
 - d) Oral first-generation antihistamines can be used temporarily to help with sleep



CASE STUDY

EK comes back to the pharmacy two weeks later. She tells you she has an upcoming appointment with her family physician because she is still struggling with the itch which is now significantly affecting her sleep, and consequently, her day-to-day functioning. Moreover, TEVA-LENOLTEC No. 3 (acetaminophen 300 mg/caffeine 15 mg/codeine 15 mg) no longer seems to work to manage her spinal pain.

2. Which of the following would be an appropriate alternative to suggest to the physician?
- a) Doxepin
 - b) Oral second-generation antihistamine (e.g. loratadine)
 - c) Gabapentin
 - d) Capsaicin



CASE STUDY

EK comes back to the pharmacy two weeks later. She tells you she has an upcoming appointment with her family physician because she is still struggling with the itch which is now significantly affecting her sleep, and consequently, her day-to-day functioning. Moreover, TEVA-LENOLTEC No. 3 (acetaminophen 300 mg/caffeine 15 mg/codeine 15 mg) no longer seems to work to manage her spinal pain.

3. What can you suggest to EK for temporary relief until her doctor's appointment?
- a) Oral or topical first-generation antihistamines can help with sleep due to their sedating effects
 - b) Since pruritus is considered a minor ailment, provide a 2-week prescription for gabapentin and fax doctor to inform
 - c) Since pruritus is considered a minor ailment, provide a 2-week prescription for butorphanol and fax doctor to inform
 - d) Oral diphenhydramine can be temporarily used for sleep until EK sees her doctor



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CHANGE LOG

March 2025

- Content reviewed; no changes made

August 2025

- Formatting and grammar changes throughout
- Slide 3: Updated legend
- Slide 5: Removed dry skin (xerosis)
- Slide 8: Removed FYI
- Slide 11: Updated points to align with Can Fam Phys pruritus guidelines
- Slide 16: Added level of evidence and footnotes
- Slides 17, 18: Added FYI
- Slide 19: Replaced hydroxyzine with diphenhydramine
- Slide 27: Updated references

October 2025

- Minor formatting and grammatical changes made
- Slide 3: Updated legend
- Slide 22: Elaborated on monitoring parameters
- Slide 23: Changed title to Takeaway

February 2026

- Slide 16: Added information on difelikefalin